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National Institute of Technology, Kurukshetra
END SEMESTER EXAMINATION Aug 2026 (EVEN)
Question Paper

B.Tech IV Semester (Artificial Intelligence and Machine Learning)

Course Name: Neural Networks
Course Code: AIPC -204
Number of Questions to be attempted: 6
Total number of questions: 6

Maximum Marks: 50
Time Allowed: 3 hours
Total Number of pages: 1

- (a) Give an example each of binarization and discretization in data preprocessing. Why do we need to perform these steps?
(b) Explain binary mask vector with the help of example and describe how is it used in DropConnect regularization. (4, 5)
- (a) What is the meaning of convolution operation? Describe mathematically. Is it same as cross correlation?
(b) What is the effect of kernel flipping? Show the proof of commutativity and associativity for convolution operation. (4, 4)
- Consider a 3×3 image matrix I (shown on left) and a 2×2 filter K for a CNN:

0	3	6
1	4	7
2	5	8

0	2
1	3

Obtain the output feature map by applying convolution, without padding (stride = 1).
Suppose, this feature map goes as input, x to a neural network

such that, $w =$

1	0	-1	2
0	1	1	-1

and $b =$

1
0

Now, obtain the final output, y from the neural network, where notations have usual meanings. (7)

- (a) What is gradient based optimization? Explain the effect of adding momentum with respect to RMSprop.
(b) With the help of equations and diagrams, explain the working of a simple RNN. Why do vanishing and exploding gradients create training difficulties in RNN? (5, 6)
- What is SVD? Explain mathematically. What are some of its applications? Explain how can it be used to obtain a low-rank approximation of a matrix. (7)
- Draw the architecture of GRU. Explain the gates, notations and corresponding mathematical equations for GRU. (8)

THE END